

Lecture 6: Properties of Fourier Series

CSE 219: Signals and Linear Systems

Anik Saha

Adjunct Lecturer, Department of CSE, BUET

Properties of Fourier Series

What happens to the coefficients when we play with the signal?

But why will we be learning these properties?

$$x(t) \longleftrightarrow a_k$$

Finding a_k directly - needs an integral.
Properties let us reuse a Fourier series we already know.

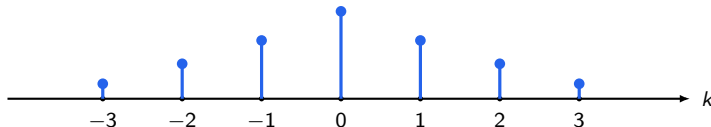
known signal $\implies$ related signal

Shift it, scale it, add it, differentiate it ... and
simply transfer the coefficients without starting from scratch.

Under the hood: every signal is a stack of simple waves

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Fourier coordinates a_k



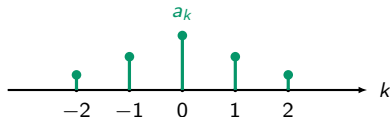
When we perform any operation on $x(t)$, we are really operating on all these simple little waves underneath.

Linearity

Linearity: adding signals simply adds their ingredients

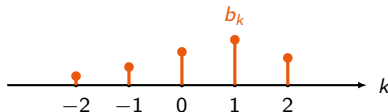
$x(t)$

$$x(t) = \sum_k a_k e^{jk\omega_0 t}$$



$y(t)$

$$y(t) = \sum_k b_k e^{jk\omega_0 t}$$



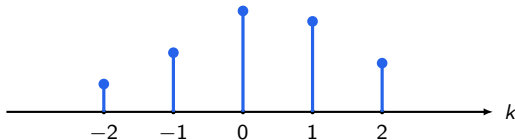
If they have the same period T , their Fourier axes line up exactly.

Same frequency meets same frequency

$$x(t) + y(t) = \sum_k a_k e^{jk\omega_0 t} + \sum_k b_k e^{jk\omega_0 t}$$

$$= \sum_k (a_k + b_k) e^{jk\omega_0 t}$$

new coefficients: $a_k + b_k$



Linearity of Fourier series

$$x(t) \longleftrightarrow a_k, \quad y(t) \longleftrightarrow b_k.$$

$$\alpha x(t) + \beta y(t) \longleftrightarrow \alpha a_k + \beta b_k.$$

mix in time $\iff$ mix the Fourier coefficients

Time Shifting

Time shift: what happens if the signal is delayed?

Goal

$$x(t) \longleftrightarrow a_k$$

$$x(t - t_0) \longleftrightarrow ?$$

Think: if we delay a signal, does its frequency change?

Simple! Just shift every simple wave underneath

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$x(t - t_0) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(t-t_0)}$$

$$= \sum_{k=-\infty}^{\infty} \underbrace{a_k e^{-jk\omega_0 t_0}}_{\text{new coefficient}} e^{jk\omega_0 t}$$

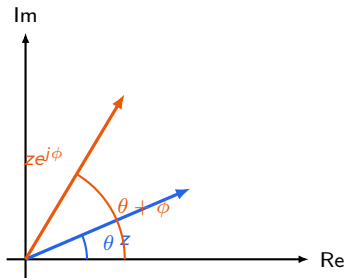
Time-shifting property

$$x(t) \longleftrightarrow a_k$$

$$x(t - t_0) \longleftrightarrow a_k e^{-jk\omega_0 t_0}$$

A delay in time becomes a phase shift in frequency.

Multiplying by $e^{j\phi}$ means rotating



Multiplication by $e^{j\phi}$ keeps the radius of rotation same,
but adds a phase-shift of ϕ to the angle.

So what does the delay factor do?

$$a_k e^{-jk\omega_0 t_0}$$

Magnitude

$$\left| e^{-jk\omega_0 t_0} \right| = 1$$

coefficient size unchanged

Phase

$$\angle e^{-jk\omega_0 t_0} = -k\omega_0 t_0$$

only phase changes

Higher-frequency components rotate more ...
because their frequency is $k\omega_0$.

Shift in time $\rightarrow$ Rotation in frequency

Delay in Time Domain

$$x(t) \rightarrow x(t - t_0)$$



(-)ve Phase Shift in Frequency Domain

$$a_k \rightarrow a_k e^{-jk\omega_0 t_0}$$



Advance in Time Domain

$$x(t) \rightarrow x(t + t_0)$$



(+)ve Phase Shift in Frequency Domain

$$a_k \rightarrow a_k e^{+jk\omega_0 t_0}$$



Problem: shift the signal, update the coefficients

Suppose

$$x(t) \longleftrightarrow a_k.$$

Find the Fourier series coefficients of

$$y(t) = x(t - 3)$$

in terms of a_k .

Answer: delay becomes a phase factor

Use the time-shifting property

$$x(t - t_0) \longleftrightarrow a_k e^{-jk\omega_0 t_0}$$

For $y(t) = x(t - 3)$, we have $t_0 = 3$. Therefore,

$$p(t) = x(t - 3) \longleftrightarrow a_k e^{-j3k\omega_0}$$

The magnitudes stay the same; only the phases rotate.

Time Reversal

Time reversal: playing the signal backward

Goal

$$x(t) \longleftrightarrow a_k$$

$$x(-t) \longleftrightarrow ?$$

Time reversal flips every simple rotating wave underneath.

Let's reverse each Fourier ingredient

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

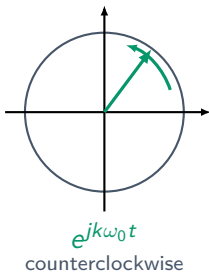
$$x(-t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(-t)} = \sum_{k=-\infty}^{\infty} a_k e^{j(-k\omega_0)t}$$

The term that used to live at $+k\omega_0$ now lives at $-k\omega_0$.

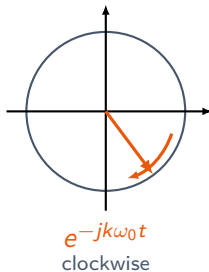
$$x(-t) \longleftrightarrow a_{-k}$$

Visual intuition: the direction of rotation simply flips!

Original wave



After $t \mapsto -t$



Imagine playing a video of the rotation backward in time.
 $+k\omega_0$ becomes $-k\omega_0$.

Time reversal property

$$x(t) \longleftrightarrow a_k$$

$$x(-t) \longleftrightarrow a_{-k}$$

Time reversal in time becomes index reversal in frequency.

Reversal in time $\rightarrow$ Reversal in frequency

Reversal in Time Domain

$$x(t) \rightarrow x(-t)$$



Reversal in Frequency Domain

$$a_k \rightarrow a_{-k}$$



A Problem on everything so far!

Suppose

$$x(t) \longleftrightarrow a_k, \quad y(t) \longleftrightarrow b_k.$$

Find the Fourier series coefficients of

$$x(4 - t) + y(t - 3)$$

in terms of a_k and b_k .

Answer: First part $\rightarrow$ Shift then Reverse

$$x(t) \longleftrightarrow a_k$$

By time shifting, $x(t + 4) \longleftrightarrow a_k e^{j4k\omega_0}$.

Now, by reversal on t , $x(-t + 4) \longleftrightarrow a_{-k} e^{-j4k\omega_0}$.

$$x(4 - t) \longleftrightarrow a_{-k} e^{-j4k\omega_0}.$$

Answer: now add the second part

$$y(t) \longleftrightarrow b_k$$

By time shifting, $y(t - 3) \longleftrightarrow a_k e^{-j3k\omega_0}$.

Therefore,

$$x(4 - t) + y(t - 3) \longleftrightarrow a_{-k} e^{-j4k\omega_0} + b_k e^{-j3k\omega_0}$$

Time Scaling

Time scaling: here we change the frequencies themselves

Goal

$$x(t) \longleftrightarrow a_k, \quad \omega_0 = \frac{2\pi}{T}$$

$$x(\alpha t) \longleftrightarrow ?$$

Time compression makes every hidden rotating wave rotate faster.

Let's scale every Fourier ingredient

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$x(\alpha t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(\alpha t)} = \sum_{k=-\infty}^{\infty} a_k e^{jk(\alpha\omega_0)t}.$$

The coefficients are still a_k ,
but the new fundamental frequency is $\omega'_0 = \alpha\omega_0$.

Two views: coefficient index vs actual frequency

Against index k

$$b_k = a_k$$

The coefficient attached to the k -th harmonic stays the same.

Against frequency ω

$$k\omega_0 \longrightarrow k\alpha\omega_0$$

The spectral lines move farther apart when $\alpha > 1$.

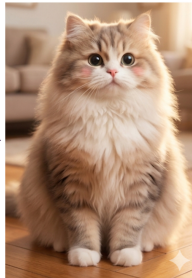
Time scaling is special: it changes the fundamental frequency itself.

(when the frequency domain is plotted against frequency values ω)

Compression in Time Domain



Expansion in Frequency Domain



Expansion in Time Domain

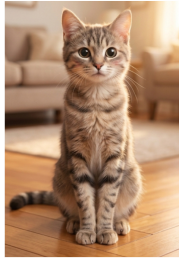


Compression in Frequency Domain



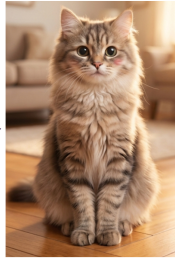
(when the frequency domain is plotted as Fourier Coefficients against Fourier indices k)

Compression in Time Domain

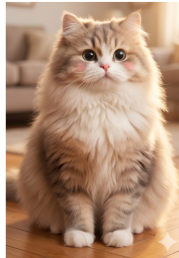


"No Change" in Fourier Coefficients

$a_k \rightarrow a_k$ (since ω_0 itself changes)

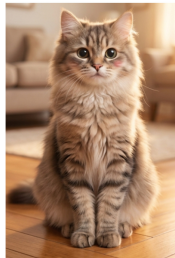


Expansion in Time Domain



"No Change" in Fourier Coefficients

$a_k \rightarrow a_k$ (since ω_0 itself changes)



Time-scaling property

$$x(t) \longleftrightarrow a_k \quad \text{with fundamental frequency } \omega_0$$

$$x(\alpha t) \longleftrightarrow a_k \quad \text{with new fundamental frequency } \alpha\omega_0$$

The coefficient sequence does not change, but the frequency ruler changes.

Practice: combine shift, reversal, and scaling

$$\text{Let, } x(t) \longleftrightarrow a_k$$

with fundamental frequency ω_0 .

Find the Fourier series coefficients of $p(t) = x(4 - 3t)$ in terms of a_k .

Think in order:

$$x(t) \longrightarrow x(t + 4) \longrightarrow x(4 - t) \longrightarrow x(4 - 3t).$$

Answer: first shift, then reverse, then scale

First shift left by 4:

$$x(t + 4) \longleftrightarrow a_k e^{j4k\omega_0}.$$

Then reverse:

$$x(4 - t) \longleftrightarrow a_{-k} e^{-j4k\omega_0}.$$

Finally scale time by 3:

$$x(4 - 3t) \longleftrightarrow a_{-k} e^{-j4k\omega_0}. \text{ (Yes, no change!)}$$

The new fundamental frequency becomes $\omega'_0 = 3\omega_0$.

Differentiation

Simple, every little wave gets differentiated!

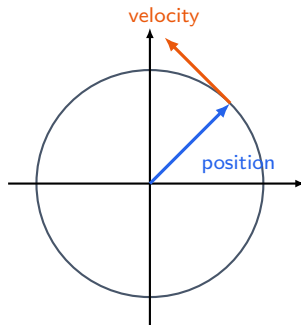
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$\frac{d}{dt}x(t) = \sum_{k=-\infty}^{\infty} a_k \frac{d}{dt}e^{jk\omega_0 t}$$

$$\frac{d}{dt}e^{jk\omega_0 t} = jk\omega_0 e^{jk\omega_0 t}$$

$$\boxed{\frac{d}{dt}x(t) \longleftrightarrow (jk\omega_0) \times a_k}$$

But why does the multiplier look like $jk\omega_0$?

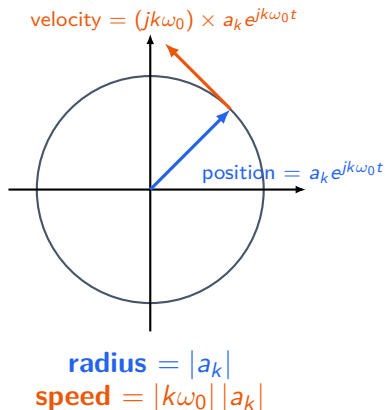


$$\begin{aligned}\text{radius} &= |a_k| \\ \text{speed} &= |k\omega_0| |a_k|\end{aligned}$$

Differentiation gives velocity.

For circular motion, speed is ωr . Here $\omega = k\omega_0$ and $r = |a_k|$.

Well! But where does the j come from?



We need to think in vectors, and therefore, direction matters!
Velocity is tangent to position, and multiplying by j does exactly that!

Problem: combine properties with differentiation

$$x(t) \longleftrightarrow a_k$$

with fundamental frequency ω_0 .

Find the Fourier series coefficients of $p(t) = \frac{d}{dt}x(4 - 3t)$ in terms of a_k .

Use the result from the previous practice:

$$x(4 - 3t) \longleftrightarrow a_{-k} e^{-j4k\omega_0}, \quad \omega'_0 = 3\omega_0.$$

Answer: differentiate using the current ω'_0

$$x(4 - 3t) \longleftrightarrow a_{-k} e^{-j4k\omega_0}, \quad \omega'_0 = 3\omega_0.$$

Now differentiate:

$$\frac{d}{dt}x(4 - 3t) \longleftrightarrow jk\omega'_0 \left(a_{-k} e^{-j4k\omega_0} \right).$$

$$\text{Since } \omega'_0 = 3\omega_0,$$

$$p(t) \longleftrightarrow j3k\omega_0 a_{-k} e^{-j4k\omega_0}$$

Let's draw derivatives of signals!

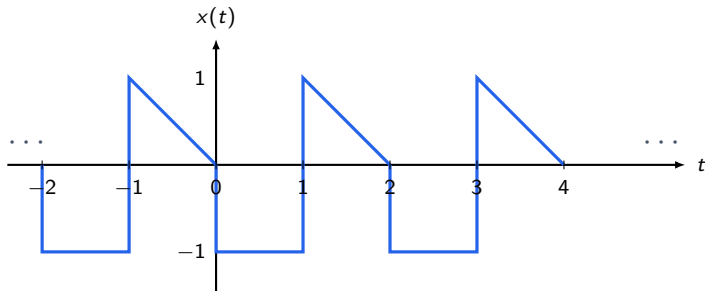
For ordinary smooth parts, derivative means slope.
At a jump, however, the derivative contains an impulse.

$$\text{jump of height } A \text{ at } t = t_0 \implies A\delta(t - t_0)$$

So the derivative has two parts:

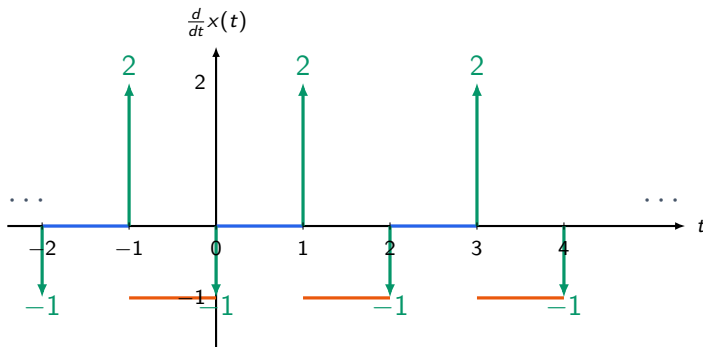
slope between jumps + impulses at jumps.

Practice: sketch the derivative



Find $\frac{d}{dt}x(t)$. Remember to include impulses at jumps.

Answer: slopes plus impulses



Flat part $\rightarrow$ slope 0 (blue). Slanted part $\rightarrow$ slope -1 (orange).
Each jump $\rightarrow$ an impulse equal to its jump height.

A little detour: impulse as a value-picker

Let's sum up the product of an arbitrary signal $x[n]$ and an impulse

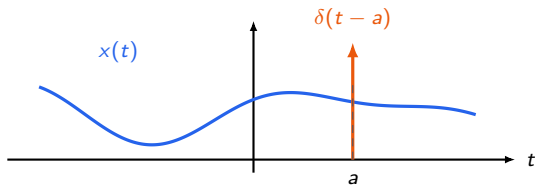
$$\sum_{n=-\infty}^{\infty} x[n]\delta[n-a]$$

$\delta[n-a]$ is zero everywhere except at $n=a$. So only one term survives.

$$\sum_{n=-\infty}^{\infty} x[n]\delta[n-a] = x[a]$$

Continuous-time works the same way

$$\int_{-\infty}^{\infty} x(t) \delta(t - a) dt$$

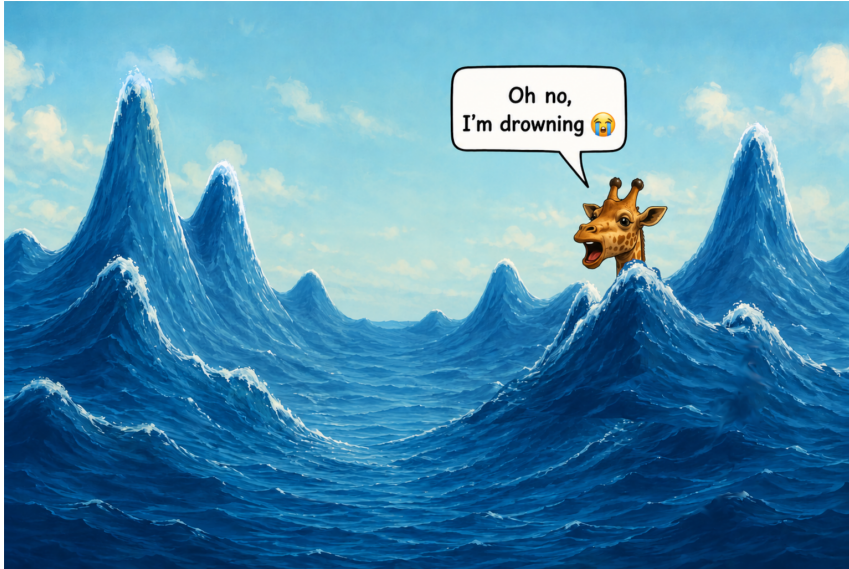


$$\int_{-\infty}^{\infty} x(t) \delta(t - a) dt = x(a)$$

A giraffe only feels the height at the exact point it stands!



A giraffe only feels the height at the exact point it stands!



One caution: the impulse must be inside the interval

$$\int_L^R x(t)\delta(t-a) dt = \begin{cases} x(a), & L < a < R, \\ 0, & a \notin [L, R]. \end{cases}$$

The impulse extracts $x(a)$
but only if the integration window actually sees the impulse.

Practice: use the sifting property

Let $x(t) = e^t$. Find:

$$(a) \int_{-\infty}^{\infty} x(t) \delta(t) dt$$

$$(b) \int_1^{\infty} x(t) \delta(t) dt$$

$$(c) \int_{-5}^5 x(t) [\delta(t+1) + 2\delta(t-2)] dt$$

Answer: each impulse picks one value

$$(a) \quad \int_{-\infty}^{\infty} e^t \delta(t) dt = e^0 = 1.$$

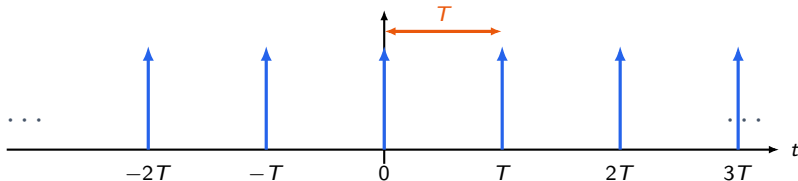
$$(b) \quad \int_1^{\infty} e^t \delta(t) dt = 0,$$

because the impulse at $t = 0$ is outside the interval.

$$(c) \quad e^{-1} + 2e^2.$$

Problem: Let's find the Fourier series of an impulse train

$$x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT)$$

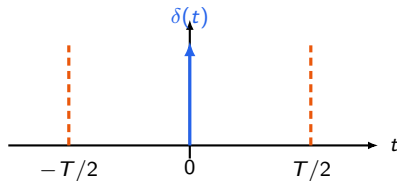


Find the complex exponential Fourier series coefficients a_k .

Choose one period that contains exactly one impulse

Choose the period

$$-\frac{T}{2} < t < \frac{T}{2}.$$



$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt.$$

Now let's use sifting to get the coefficients

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt$$

By sifting, the impulse at $t = 0$ picks the value of $e^{-jk\omega_0 t}$ at $t = 0$.

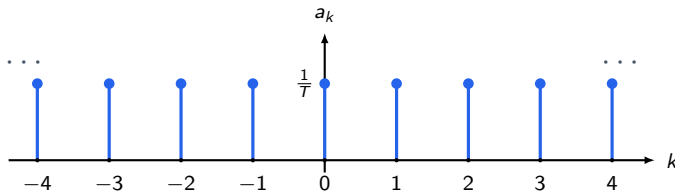
$$a_k = \frac{1}{T} e^{-jk\omega_0 \cdot 0} = \boxed{\frac{1}{T}}.$$

Every Fourier coefficient has the same value:

$$a_k = \frac{1}{T}, \quad k = 0, \pm 1, \pm 2, \dots$$

Frequency-domain picture

$$x(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT) \iff a_k = \frac{1}{T}.$$



The coefficient sequence is flat.
As a line spectrum in ω , these lines sit at $\omega = k\omega_0$.

A train remains a train!

TIME DOMAIN

A train



Fourier Series

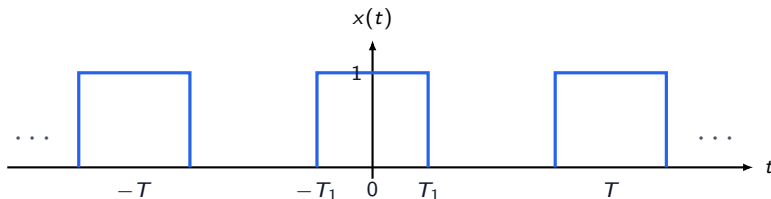
FREQUENCY DOMAIN

Still a train



Problem: differentiate a periodic rectangular pulse

$$x(t) = \begin{cases} 1, & |t| < T_1, \\ 0, & T_1 < |t| < T/2, \end{cases} \quad x(t + T) = x(t).$$



Differentiate $x(t)$, find the Fourier series coefficients of $\frac{d}{dt}x(t)$, and match with the differentiation property.

Known Fourier series of the original signal

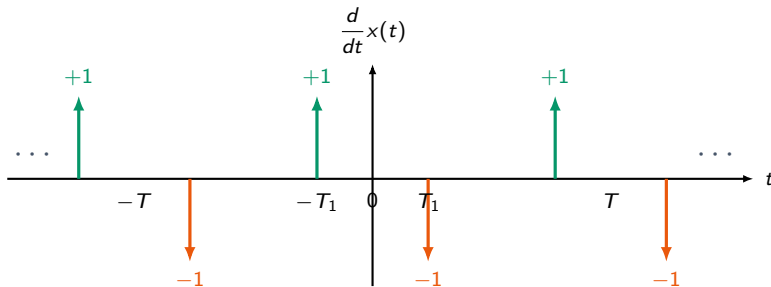
From previous slides, for the rectangular pulse,

$$x(t) \longleftrightarrow a_k.$$

$$a_0 = \frac{2T_1}{T}$$

$$a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0, \quad \omega_0 = \frac{2\pi}{T}.$$

What does the derivative look like?



Rising edge gives $+\delta$. Falling edge gives $-\delta$.

Write the derivative using impulses

In one period,

$$\frac{d}{dt}x(t) = \delta(t + T_1) - \delta(t - T_1).$$

For the periodic signal,

$$\frac{d}{dt}x(t) = \sum_{m=-\infty}^{\infty} [\delta(t + T_1 - mT) - \delta(t - T_1 - mT)].$$

Now find its Fourier coefficients directly using sifting.

Let's first find the coefficients directly

Let the coefficients of $\frac{d}{dt}x(t)$ be b_k . Then

$$b_k = \frac{1}{T} \int_{-T/2}^{T/2} [\delta(t + T_1) - \delta(t - T_1)] e^{-jk\omega_0 t} dt.$$

Using sifting,

$$\begin{aligned} b_k &= \frac{1}{T} [e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}] \\ &= \boxed{\frac{2j}{T} \sin(k\omega_0 T_1)}. \end{aligned}$$

Now match with the differentiation property

Differentiation property says, $\frac{d}{dt}x(t) \longleftrightarrow jk\omega_0 a_k$.

$$\text{For } k \neq 0, \quad a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}.$$

$$jk\omega_0 a_k = jk\omega_0 \frac{\sin(k\omega_0 T_1)}{k\pi} = j \frac{\omega_0}{\pi} \sin(k\omega_0 T_1).$$

$$\text{Since } \omega_0 = \frac{2\pi}{T}, \quad jk\omega_0 a_k = \boxed{\frac{2j}{T} \sin(k\omega_0 T_1)}.$$

Perfect match

Direct calculation gave

$$b_k = \frac{2j}{T} \sin(k\omega_0 T_1). \text{ (this already also gives } b_0 = 0)$$

Differentiation property gave

$$jk\omega_0 a_k = \frac{2j}{T} \sin(k\omega_0 T_1).$$

$$b_k = (jk\omega_0) \times a_k$$

Integration

Integration: reverse of differentiation

$$x(t) \longleftrightarrow a_k$$

Differentiation says, $\frac{d}{dt}x(t) \longleftrightarrow jk\omega_0 a_k$.

So integration should divide by the same factor:

$$\int x(t) dt \longleftrightarrow \frac{a_k}{jk\omega_0}, \quad k \neq 0.$$

One important warning: the constant term

Integration cannot recover the constant term.

Why?

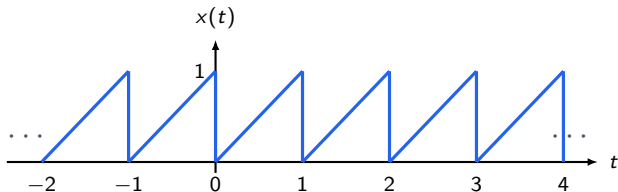
$$\frac{d}{dt}(x(t) + C) = \frac{d}{dt}x(t).$$

So after integrating, the a_0 term must be determined separately.

$$b_k = \frac{a_k}{jk\omega_0}, \quad k \neq 0$$

Problem: use differentiation first, then integrate back

$$x(t) = t, \quad 0 \leq t < 1, \quad x(t+1) = x(t).$$

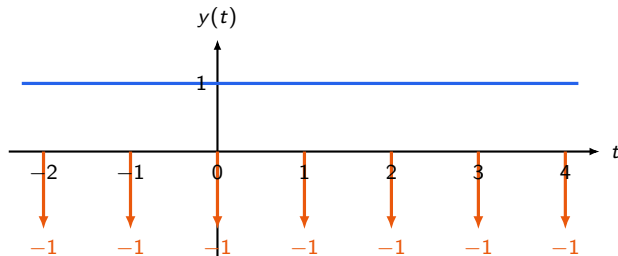


Find the Fourier series coefficients of $x(t)$
by first finding the series for the derivative.

Derivative of the sawtooth

Between jumps, the slope is 1. At each integer, the signal jumps down by 1.

$$y(t) = \frac{d}{dt}x(t) = 1 - \sum_{m=-\infty}^{\infty} \delta(t - m).$$



Fourier coefficients of the derivative

$$y(t) = 1 - \sum_{m=-\infty}^{\infty} \delta(t - m).$$

The constant part 1 has coefficients

$$1 \longleftrightarrow \begin{cases} 1, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

Fourier coefficients of the derivative

$$y(t) = 1 - \sum_{m=-\infty}^{\infty} \delta(t - m).$$

The impulse train part has coefficients 1 for all k . ($\frac{1}{T} = \frac{1}{1} = 1$)

$$\text{Therefore, } y(t) \longleftrightarrow b_k = \begin{cases} 0, & k = 0, \\ -1, & k \neq 0. \end{cases}$$

Integrate back to get the coefficients of $x(t)$

$$y(t) = \frac{d}{dt}x(t) \implies b_k = jk\omega_0 a_k.$$

So, for $k \neq 0$,

$$a_k = \frac{b_k}{jk\omega_0}.$$

Here $b_k = -1$ and $\omega_0 = 2\pi$, so

$$a_k = \frac{-1}{j2\pi k} = \frac{j}{2\pi k}, \quad k \neq 0.$$

Do not forget a_0

Integration did not determine a_0 .

Let's just find it from the average value:

$$a_0 = \int_0^1 t \, dt = \frac{1}{2}.$$

$$a_0 = \frac{1}{2}, \quad a_k = \frac{j}{2\pi k}, \quad k \neq 0.$$

Multiplication

New goal: what about multiplying two signals?

So far, if

$$x(t) \longleftrightarrow a_k, \quad y(t) \longleftrightarrow b_k,$$

then linearity gives us

$$x(t) + y(t) \longleftrightarrow a_k + b_k.$$

But what about multiplication?

$$x(t)y(t) \longleftrightarrow ?$$

Simple! Let's multiply the two Fourier series

$$x(t) = \sum_{p=-\infty}^{\infty} a_p e^{jp\omega_0 t}, \quad y(t) = \sum_{q=-\infty}^{\infty} b_q e^{jq\omega_0 t}.$$

$$x(t)y(t) = \left(\sum_{p=-\infty}^{\infty} a_p e^{jp\omega_0 t} \right) \left(\sum_{q=-\infty}^{\infty} b_q e^{jq\omega_0 t} \right).$$

We use p and q because the two sums need separate indices.

This is nothing but polynomial multiplication!

Let, $z = e^{j\omega_0 t}$. Then,

$$e^{jp\omega_0 t} = z^p, \quad e^{jq\omega_0 t} = z^q.$$

$$x(t) = \sum_p a_p z^p, \quad y(t) = \sum_q b_q z^q.$$

It becomes becomes multiplying two coefficient-polynomials in z !

Which terms create the k -th frequency?

When we multiply one term from $x(t)$ and one term from $y(t)$,

$$a_p e^{jp\omega_0 t} \cdot b_q e^{jq\omega_0 t} = a_p b_q e^{j(p+q)\omega_0 t}.$$

To contribute to the k -th harmonic, we need

$$p + q = k.$$

So $q = k - p$, and the k -th coefficient becomes

$$c_k = \sum_{p=-\infty}^{\infty} a_p b_{k-p}.$$

Multiplication property: Convolution Returns, Again!!!

$$x(t) \longleftrightarrow a_k, \quad y(t) \longleftrightarrow b_k,$$

$$x(t)y(t) \longleftrightarrow c_k = (a * b)_k$$

$$c_k = \sum_{p=-\infty}^{\infty} a_p b_{k-p}.$$

Multiplication in time becomes convolution in frequency.

Practice: square a simple signal

Let, $x(t) = 1 + \cos(\omega_0 t)$.

Its nonzero Fourier coefficients are

$$a_{-1} = \frac{1}{2}, \quad a_0 = 1, \quad a_1 = \frac{1}{2},$$

Find the Fourier series coefficients of

$$p(t) = x^2(t).$$

Answer: convolve the coefficient sequence with itself

Since, $p(t) = x(t)x(t)$,

$$c_k = (a * a)_k = \sum_p a_p a_{k-p}.$$

Only a_{-1}, a_0, a_1 are nonzero. So we compute the convolution:

k	-2	-1	0	1	2
c_k	$\frac{1}{4}$	1	$\frac{3}{2}$	1	$\frac{1}{4}$

Let's check using ordinary trigonometry

$$x^2(t) = (1 + \cos(\omega_0 t))^2$$

$$= 1 + 2 \cos(\omega_0 t) + \cos^2(\omega_0 t)$$

$$= \frac{3}{2} + 2 \cos(\omega_0 t) + \frac{1}{2} \cos(2\omega_0 t).$$

$$\text{Therefore, } c_0 = \frac{3}{2}, \quad c_{\pm 1} = 1, \quad c_{\pm 2} = \frac{1}{4}.$$

Thank you!